

BankTransaction Ledger SQL:

create database bank;

Query OK, 1 row affected (0.003 sec)

MariaDB [(none)]> use bank;

Database changed

▪ **Customers Table:**

MariaDB [bank]> CREATE TABLE Customers (

-> CustomerID INT PRIMARY KEY,

-> CustomerName VARCHAR(100),

-> Phone VARCHAR(15),

-> Address VARCHAR(200)

->);

Query OK, 0 rows affected (0.029 sec)

MariaDB [bank]> INSERT INTO Customers

-> VALUES

-> (1, 'Ravi Kumar', '9876543210', 'Hyderabad'),

-> (2, 'Anjali Sharma', '9123456780', 'Vijayawada');

Query OK, 2 rows affected (0.004 sec)

Records: 2 Duplicates: 0 Warnings: 0

MariaDB [bank]> insert into Customers
values(3,'Rakshitha','9951349223','AndhraPradesh');

Query OK, 1 row affected (0.003 sec)

MariaDB [bank]> select*from Customers;

```
+-----+-----+-----+-----+
| CustomerID | CustomerName | Phone    | Address    |
+-----+-----+-----+-----+
|      1 | Ravi Kumar   | 9876543210 | Hyderabad  |
|      2 | Anjali Sharma | 9123456780 | Vijayawada  |
|      3 | Rakshitha    | 9951349223 | AndhraPradesh |
+-----+-----+-----+-----+
```

3 rows in set (0.001 sec)

▪ **Accounts Table:**

CREATE TABLE Accounts (

- > AccountID INT PRIMARY KEY,
- > CustomerID INT,
- > AccountType VARCHAR(20),
- > Balance DECIMAL(12,2),
- > FOREIGN KEY (CustomerID) REFERENCES Customers(CustomerID)
- >);

Query OK, 0 rows affected (0.054 sec)

MariaDB [bank]> INSERT INTO Accounts

-> VALUES

-> (101, 1, 'Savings', 50000),

-> (102, 2, 'Current', 30000);

Query OK, 2 rows affected (0.005 sec)

Records: 2 Duplicates: 0 Warnings: 0

MariaDB [bank]> insert into Accounts values(103, 3,'Savings',10000);

Query OK, 1 row affected (0.020 sec)

*MariaDB [bank]> select*from Accounts;*

+-----+-----+-----+-----+

| AccountID | CustomerID | AccountType | Balance |

+-----+-----+-----+-----+

| 101 | 1 | Savings | 50000.00 |

| 102 | 2 | Current | 30000.00 |

| 103 | 3 | Savings | 10000.00 |

+-----+-----+-----+-----+

3 rows in set (0.001 sec)

▪ Transactions Table:

CREATE TABLE Transactions (

-> TransactionID INT PRIMARY KEY,

- > AccountID INT,
- > TransactionDate DATETIME,
- > TransactionType VARCHAR(20),
- > Amount DECIMAL(12,2),
- > Description VARCHAR(200),
- > FOREIGN KEY (AccountID) REFERENCES Accounts(AccountID)
- >);

Query OK, 0 rows affected (0.069 sec)

MariaDB [bank]> INSERT INTO Transactions

-> VALUES

-> (1001, 101, '2026-06-20 09:00:00', 'Deposit', 10000, 'Cash Deposit'),

-> (1002, 101, '2026-06-20 11:00:00', 'Withdrawal', 5000, 'ATM Withdrawal'),

-> (1003, 102, '2026-06-20 12:00:00', 'Deposit', 15000, 'Online Transfer');

Query OK, 3 rows affected (0.023 sec)

Records: 3 Duplicates: 0 Warnings: 0

MariaDB [bank]> select*from Transactions;

```
+-----+-----+-----+-----+-----+-----+
-----+
```

```
| TransactionID | AccountID | TransactionDate | TransactionType |
Amount | Description |
```

```

+-----+-----+-----+-----+-----+-----+
-----+
|      1001 |      101 | 2026-06-20 09:00:00 | Deposit      | 10000.00 |
Cash Deposit |
|      1002 |      101 | 2026-06-20 11:00:00 | Withdrawal   | 5000.00  |
| ATM Withdrawal |
|      1003 |      102 | 2026-06-20 12:00:00 | Deposit      | 15000.00 |
Online Transfer |
+-----+-----+-----+-----+-----+-----+
-----+

```

3 rows in set (0.001 sec)

▪ **View Transaction Ledger:**

SELECT

```

-> t.TransactionID,
-> c.CustomerName,
-> a.AccountID,
-> t.TransactionDate,
-> t.TransactionType,
-> t.Amount,
-> t.Description
-> FROM Transactions t
-> JOIN Accounts a ON t.AccountID = a.AccountID
-> JOIN Customers c ON a.CustomerID = c.CustomerID
-> ORDER BY t.TransactionDate;

```

```

+-----+-----+-----+-----+-----+-----+
----+-----+

| TransactionID | CustomerName | AccountID | TransactionDate |
TransactionType | Amount | Description |
+-----+-----+-----+-----+-----+-----+
----+-----+

|      1001 | Ravi Kumar |      101 | 2026-06-20 09:00:00 | Deposit
| 10000.00 | Cash Deposit |
|      1002 | Ravi Kumar |      101 | 2026-06-20 11:00:00 |
Withdrawal | 5000.00 | ATM Withdrawal |
|      1003 | Anjali Sharma |      102 | 2026-06-20 12:00:00 | Deposit
| 15000.00 | Online Transfer |

```

```

+-----+-----+-----+-----+-----+-----+
----+-----+

```

3 rows in set (0.002 sec)

▪ **ID 101 STATEMENT:**

SELECT

- > TransactionDate,
- > TransactionType,
- > Amount,
- > Description
- > FROM Transactions
- > WHERE AccountID = 101
- > ORDER BY TransactionDate;

```

+-----+-----+-----+-----+

```

TransactionDate	TransactionType	Amount	Description
2026-06-20 09:00:00	Deposit	10000.00	Cash Deposit
2026-06-20 11:00:00	Withdrawal	5000.00	ATM Withdrawal

2 rows in set (0.001 sec)

▪ **Account Balance of ID 101&102:**

```

SELECT
  -> AccountID,
  -> SUM(
  ->   CASE
  ->     WHEN TransactionType = 'Deposit' THEN Amount
  ->     WHEN TransactionType = 'Withdrawal' THEN -Amount
  ->     ELSE 0
  ->   END
  -> ) AS NetBalance
  -> FROM Transactions
  -> GROUP BY AccountID;

```

AccountID	NetBalance
101	5000.00
102	15000.00

2 rows in set (0.001 sec)